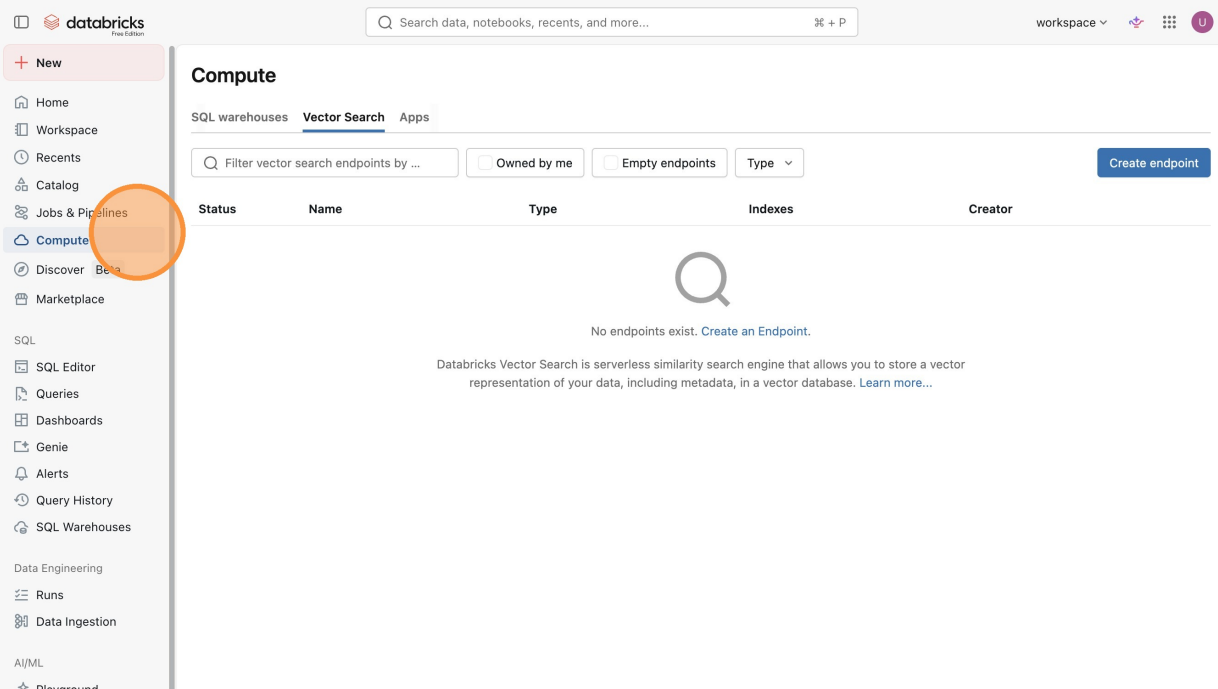


Creating a Vector Search Index in Databricks

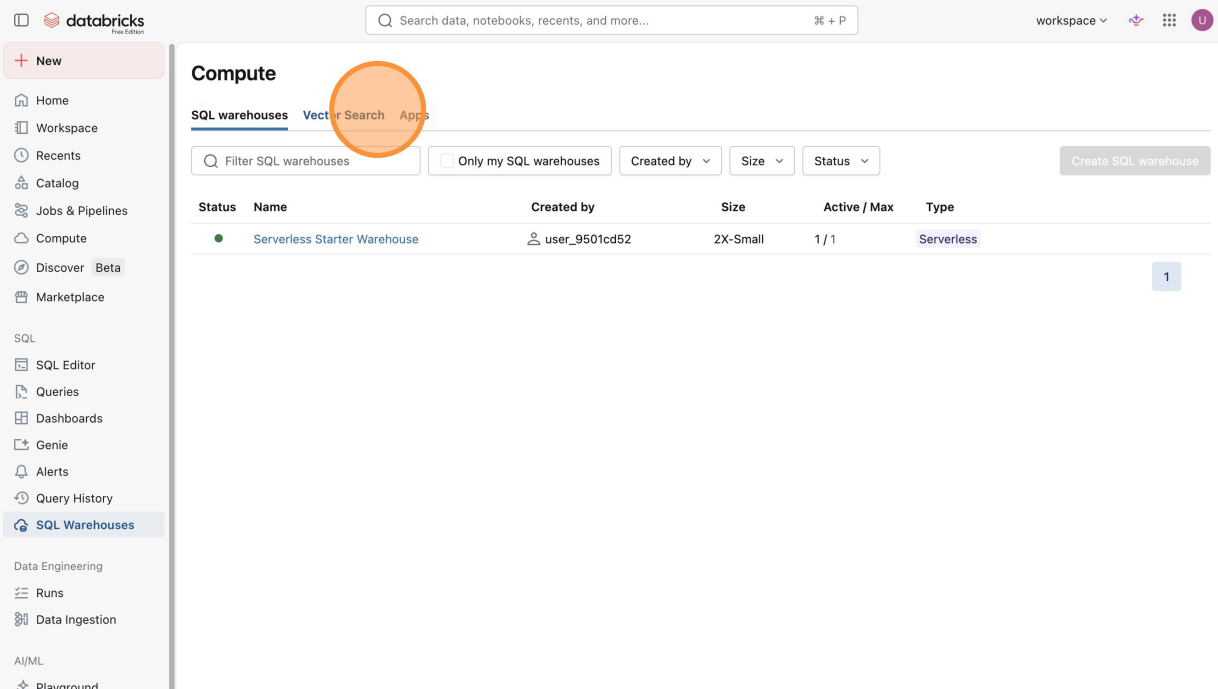
Learn how to set up a vector search endpoint and index to enable advanced semantic search capabilities on your data. This guide walks you through the configuration process to link your specific tables and embedding models effectively.

1 Click "Compute"



The screenshot shows the Databricks web interface. On the left sidebar, the 'Compute' menu item is highlighted with an orange circle. The main content area is titled 'Compute' and has tabs for 'SQL warehouses', 'Vector Search', and 'Apps'. The 'Vector Search' tab is active. Below the tabs, there is a search bar and filters for 'Owned by me', 'Empty endpoints', and 'Type'. A 'Create endpoint' button is in the top right. The main area displays a table with columns: Status, Name, Type, Indexes, and Creator. The table is empty, and a large magnifying glass icon is centered over it. Below the table, a message states: 'No endpoints exist. [Create an Endpoint.](#) Databricks Vector Search is serverless similarity search engine that allows you to store a vector representation of your data, including metadata, in a vector database. [Learn more...](#)'

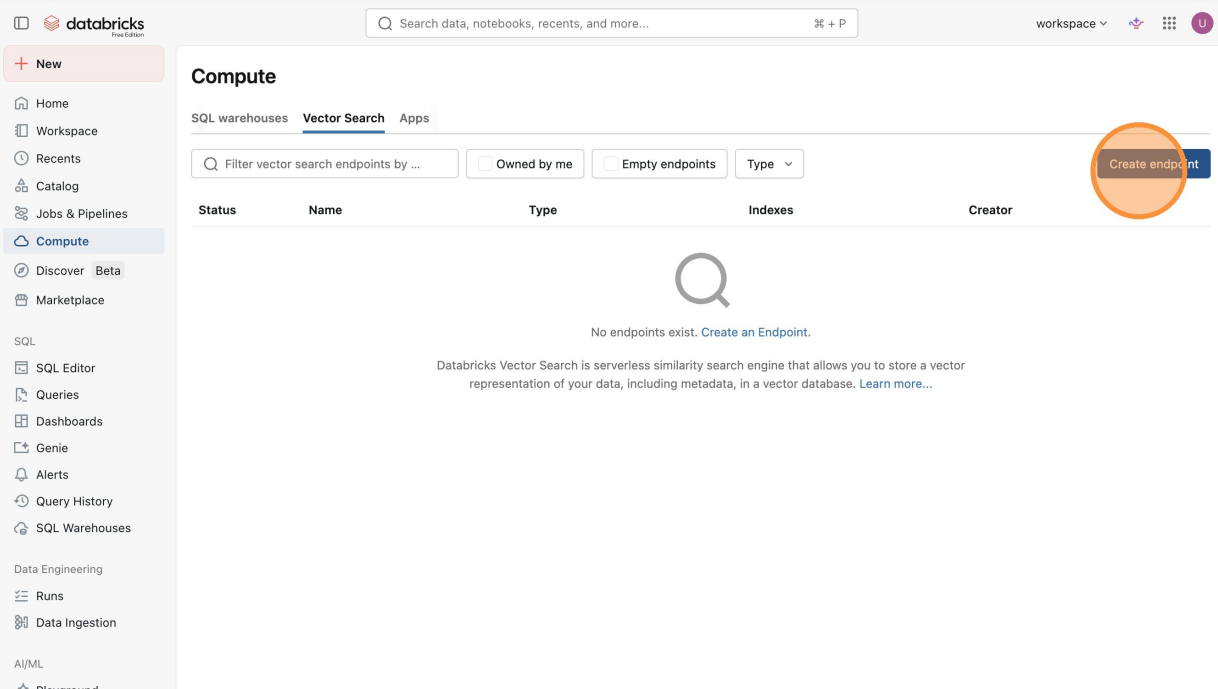
2 Click "Vector Search"



The screenshot shows the Databricks Compute page. The 'Vector Search' tab is highlighted with an orange circle. The page displays a table of SQL warehouses. The 'Serverless Starter Warehouse' is listed with a status of 'Active' and a size of '2X-Small'. The 'Create SQL warehouse' button is visible in the top right corner.

Status	Name	Created by	Size	Active / Max	Type
●	Serverless Starter Warehouse	user_9501cd52	2X-Small	1 / 1	Serverless

3 Click "Create endpoint"

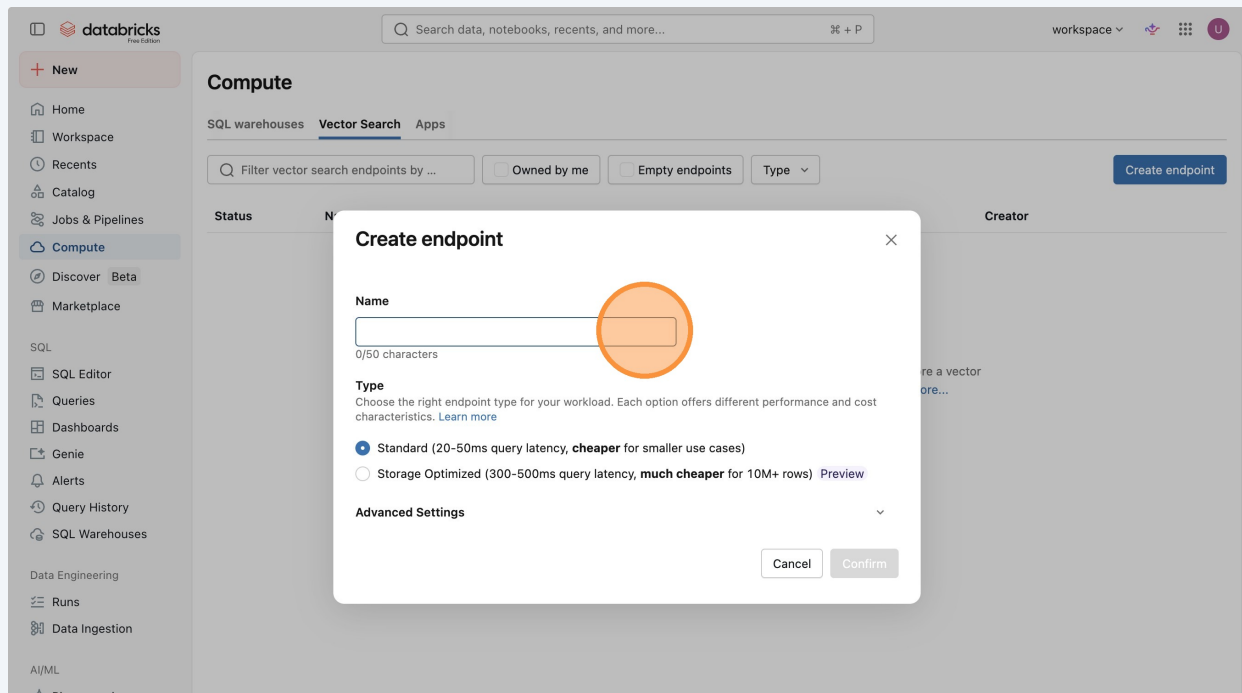


The screenshot shows the Databricks Compute page with the 'Vector Search' tab selected. The 'Create endpoint' button is highlighted with an orange circle. The page displays a message: 'No endpoints exist. Create an Endpoint.' Below this message, there is a description of Databricks Vector Search and a link to 'Learn more...'.

No endpoints exist. [Create an Endpoint.](#)

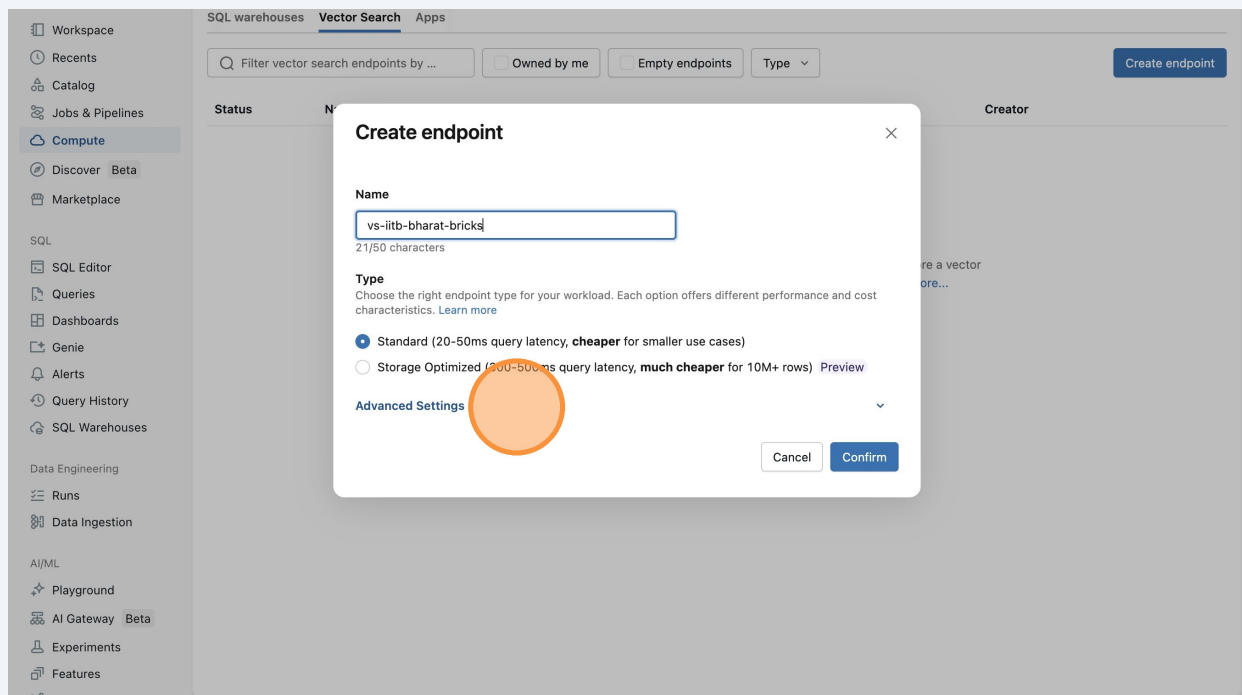
Databricks Vector Search is serverless similarity search engine that allows you to store a vector representation of your data, including metadata, in a vector database. [Learn more...](#)

4 Click the "Name" field.

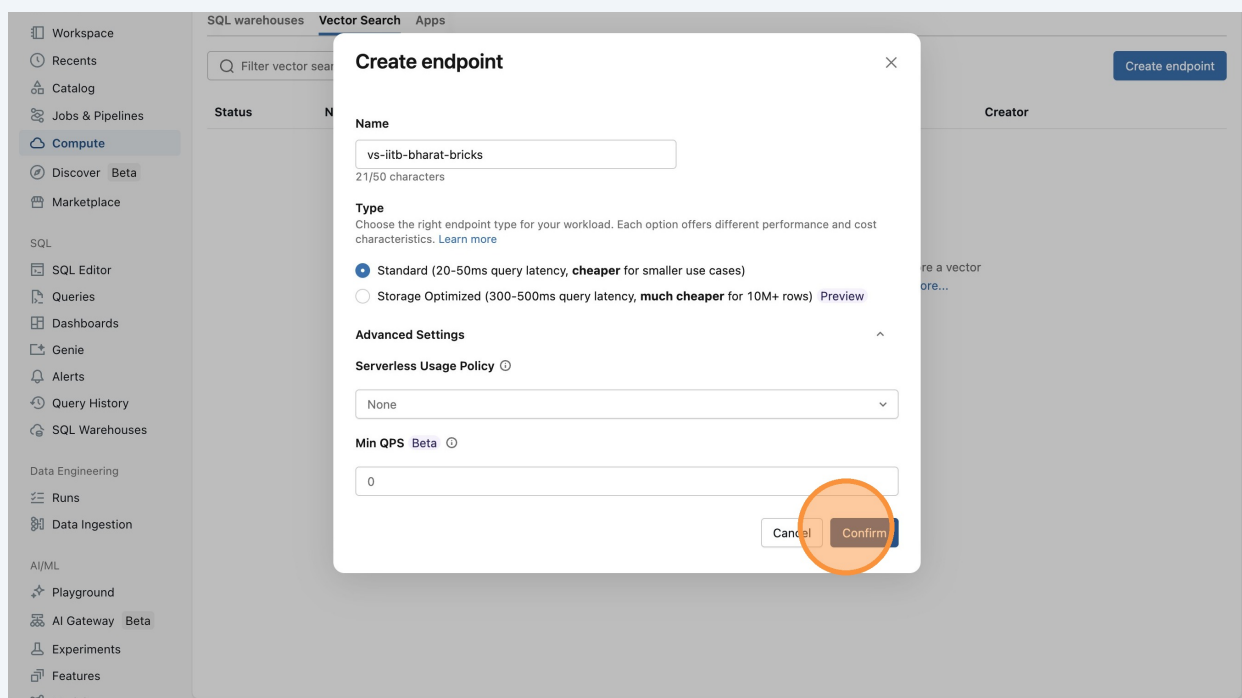


5 Type "vs-iitb-bharat-bricks"

6 Click "Advanced Settings"



7 Click "Confirm"



8 Click "bharatbricksiitb"

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Home, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, and Playground. The main area is titled 'Workspace > Users > visheshh.arya87@gmail.com'. It displays a table of items in the workspace. The 'bharatbricksiitb' Git folder is highlighted with an orange circle. The table has columns for Name, Type, Owner, and Created at.

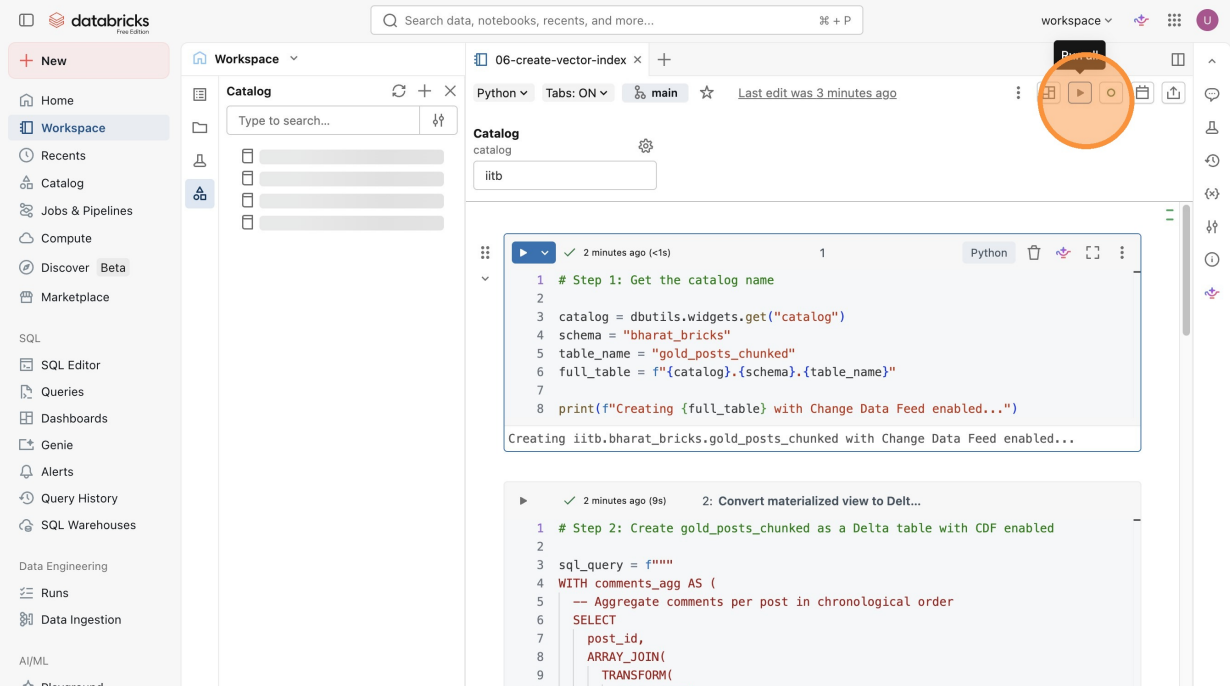
Name	Type	Owner	Created at
Drafts	Folder	user_9501cd52	Mar 28, 2026, 03...
bakehouse	Folder	user_9501cd52	Sep 04, 2025, 10...
bharatbricksiitb	Git folder	user_9501cd52	Apr 10, 2026, 10:...
secrets	File	user_9501cd52	Oct 24, 2025, 08...

9 Click "06-create-vector-index"

The screenshot shows the Databricks workspace interface. On the left is a sidebar with navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover (Beta), Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, Playground, AI Gateway (Beta), Experiments, and Features. The main area is titled 'Workspace > Shared with me > Workspace'. It displays a table of items in the workspace. The '06-create-vector-index' notebook is highlighted with an orange circle. The table has columns for Name, Type, Owner, and Created at.

Name	Type	Owner	Created at
.databricks	Folder	user_9501cd52	Apr 10, 2026, 10:...
02-data-transformation	Folder	user_9501cd52	Apr 10, 2026, 10:...
05-iitb-junta-analytics-genie	Folder	user_9501cd52	Apr 10, 2026, 10:...
07-iitb-baap-agent	Folder	user_9501cd52	Apr 10, 2026, 10:...
instructions	Folder	user_9501cd52	Apr 10, 2026, 10:...
raw_data	Folder	user_9501cd52	Apr 10, 2026, 10:...
.gitignore	File	user_9501cd52	Apr 10, 2026, 10:...
01-data-ingestion	Notebook	user_9501cd52	Apr 10, 2026, 10:...
03-metric-view	Notebook	user_9501cd52	Apr 10, 2026, 10:...
04-iitb-student-life-explorer	Dashboard	user_9501cd52	Apr 10, 2026, 10:...
05-iitb-junta-analytics	Genie space	user_9501cd52	Apr 11, 2026, 03:...
06-create-vector-index	Notebook	user_9501cd52	Apr 11, 2026, 03:...
README.md	File	user_9501cd52	Apr 10, 2026, 10:...

10 Click this button.



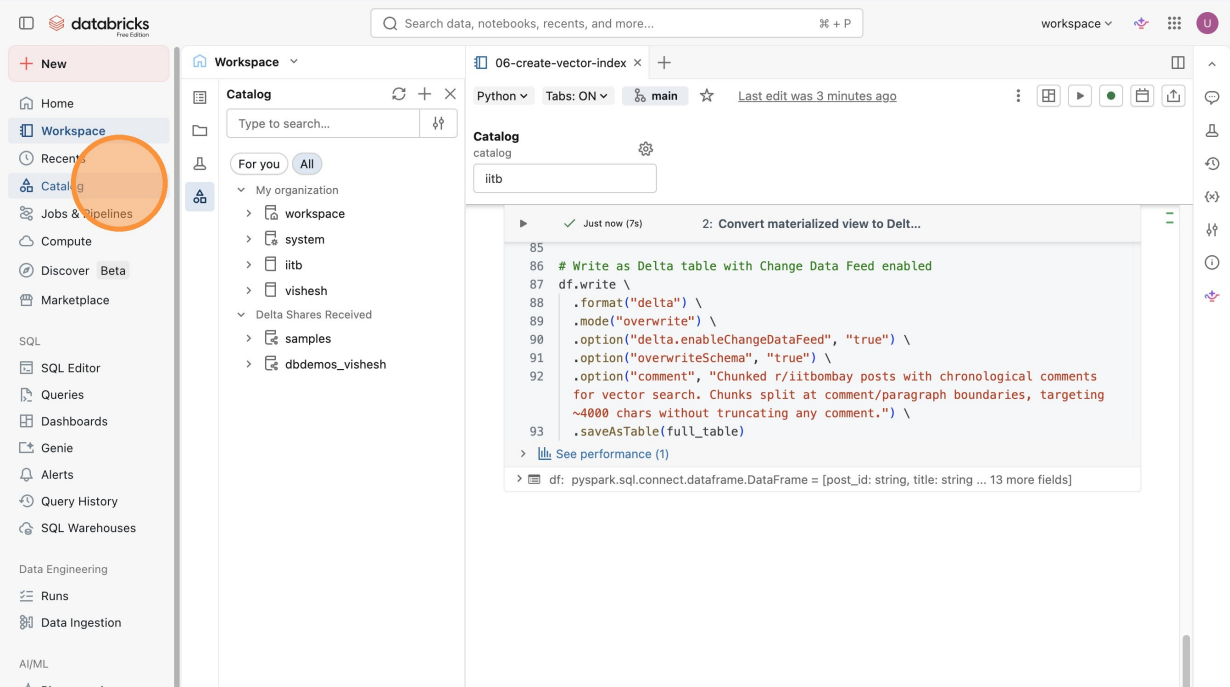
The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Workspace' tab is selected. In the main area, a notebook titled '06-create-vector-index' is open. The notebook contains two code blocks. The first code block is highlighted with a blue border and contains the following Python code:

```
1 # Step 1: Get the catalog name
2
3 catalog = dbutils.widgets.get("catalog")
4 schema = "bharat_bricks"
5 table_name = "gold_posts_chunked"
6 full_table = f"{catalog}.{schema}.{table_name}"
7
8 print(f"Creating {full_table} with Change Data Feed enabled...")
```

The output of the first code block is: 'Creating iitb.bharat_bricks.gold_posts_chunked with Change Data Feed enabled...'. The second code block is titled '2: Convert materialized view to Delt...' and contains the following SQL code:

```
1 # Step 2: Create gold_posts_chunked as a Delta table with CDF enabled
2
3 sql_query = f"""
4 WITH comments_agg AS (
5 -- Aggregate comments per post in chronological order
6 SELECT
7     post_id,
8     ARRAY_JOIN(
9     TRANSFORM(
```

11 Click "Catalog"



The screenshot shows the Databricks workspace interface. On the left sidebar, the 'Catalog' button is highlighted with an orange circle. The main area shows the 'Catalog' view, which displays a tree structure of catalogs. The 'iitb' catalog is selected, and its contents are listed below:

- workspace
- system
- iitb
- vishesh

The 'iitb' catalog is expanded, showing the following contents:

- samples
- dbdemos_vishesh

The notebook titled '06-create-vector-index' is still open, showing the second code block with the following SQL code:

```
85
86 # Write as Delta table with Change Data Feed enabled
87 df.write \
88     .format("delta") \
89     .mode("overwrite") \
90     .option("delta.enableChangeDataFeed", "true") \
91     .option("overwriteSchema", "true") \
92     .option("comment", "Chunked r/iitbomay posts with chronological comments
93     for vector search. Chunks split at comment/paragraph boundaries, targeting
94     ~4000 chars without truncating any comment.") \
95     .saveAsTable(full_table)
```

12 Click "gold_posts_chunked"

The screenshot shows the Databricks Catalog interface. On the left sidebar, the 'Catalog' section is expanded, and the 'gold_posts_chunked' table is highlighted. The main panel displays a list of suggested tables with columns: Name, Reason for suggestion, and Type. The table 'gold_posts_chunked' is listed with the reason 'You view frequently' and type 'Materialized view'.

Name	Reason for suggestion	Type
gold_comments iitb.bharat_bricks	You view frequently	Materialized view
gold_posts iitb.bharat_bricks	You view frequently	Materialized view
bharat_bricks iitb	You view frequently	Schema
iitb	You view frequently	Catalog
data	You view frequently	Volume
gold_posts_chunked iitb.bharat_bricks	You view frequently	Catalog
iitb.bharat_bricks	You viewed · 17 hours ago	Metric view
posts iitb.bharat_bricks	You viewed · 19 hours ago	Table
comments iitb.bharat_bricks	You viewed · 19 hours ago	Table
data vishesh.default	You viewed · 22 hours ago	Volume
default iitb	You viewed · 19 hours ago	Schema

13 Click "Create"

The screenshot shows the Databricks Catalog Explorer interface for the 'gold_posts_chunked' table. The 'Create' button is highlighted in the top right corner. The main panel displays the table's description and a list of columns with their types and comments.

Description

Chunked r/iitbombay posts with chronological comments for vector search. Chunks split at comment/paragraph boundaries, targeting ~4000 chars without truncating any comment.

Column	Type	Comment	Tags	Column masking
post_id	string	Unique Reddit post identifier (e.g. 1id41gh). Primary key.		
title	string	Title/headline of the Reddit post as written by the author.		
author	string	Reddit username of the post author. Set to [deleted] when the account or post was removed.		
created_at	timestamp	Timestamp (UTC) when the post was submitted. Converted from Unix epoch.		

14 Click "Vector search index"

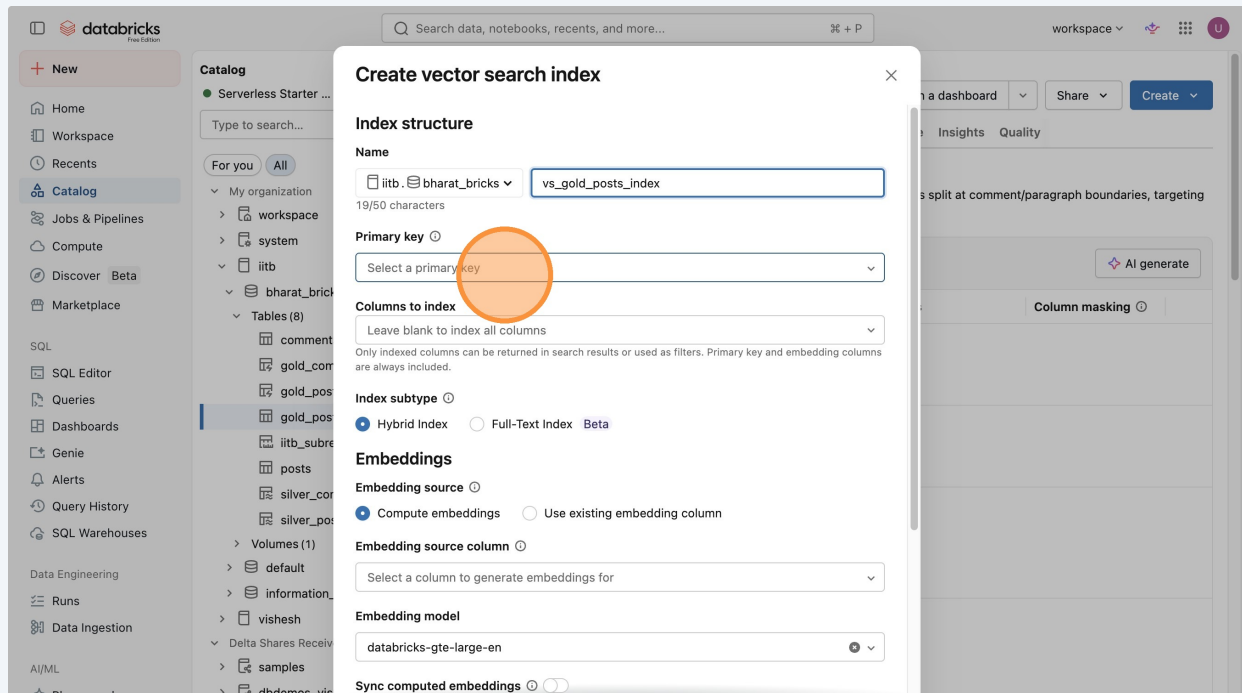
The screenshot shows the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar is visible, showing a tree structure of tables. The 'gold_posts_chunked' table is selected. On the right, the 'Overview' tab is active, displaying a table with columns: post_id, title, author, and created_at. A dropdown menu is open on the right side of the interface, showing options like 'Query', 'Notebook', 'Dashboard', 'Genie space', 'Vector search index', and 'Metric view'. The 'Vector search index' option is highlighted with an orange circle.

15 Click the "Name" field.

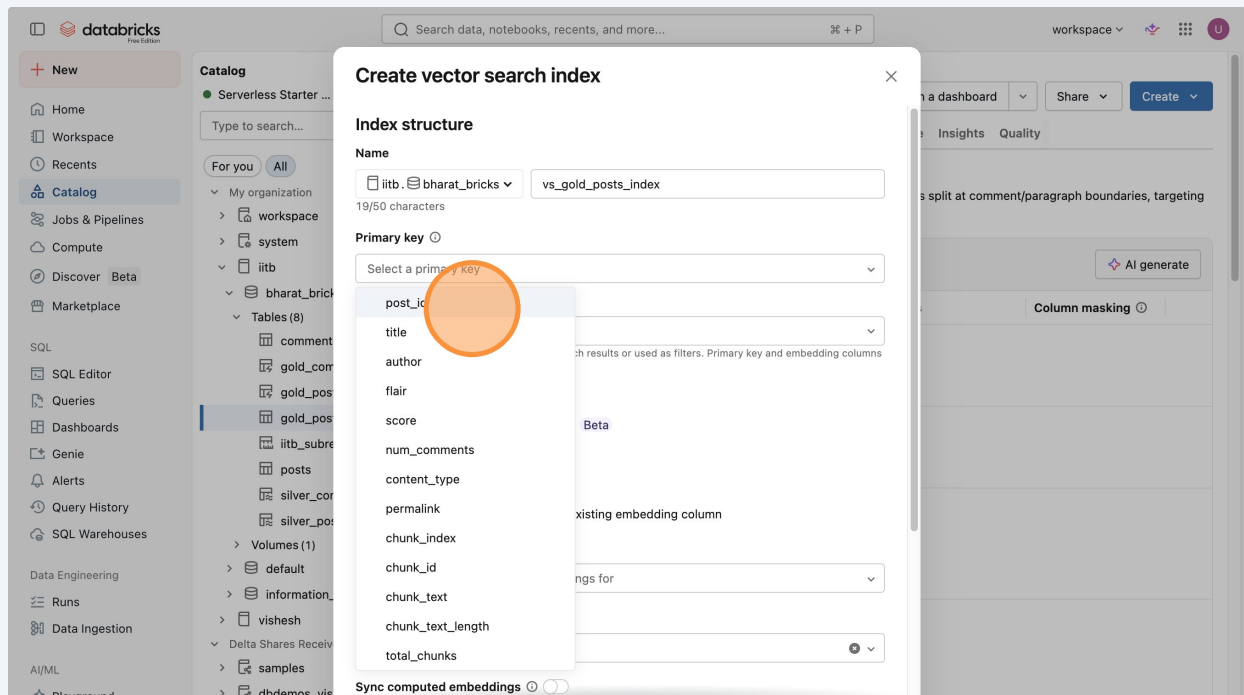
The screenshot shows the 'Create vector search index' dialog box in Databricks. The dialog has several sections: 'Index structure', 'Primary key', 'Columns to index', 'Index subtype', 'Embeddings', and 'Embedding model'. The 'Name' field under 'Index structure' is highlighted with an orange circle. The 'Name' field is currently empty, and the text 'Enter index name' is visible. The 'Primary key' field is set to 'Select a primary key'. The 'Columns to index' field is set to 'Leave blank to index all columns'. The 'Index subtype' is set to 'Hybrid Index'. The 'Embeddings' section has 'Compute embeddings' selected. The 'Embedding source column' is set to 'Select a column to generate embeddings for'. The 'Embedding model' is set to 'databricks-gte-large-en'. The 'Sync computed embeddings' checkbox is checked.

16 Type "vs_gold_posts_index"

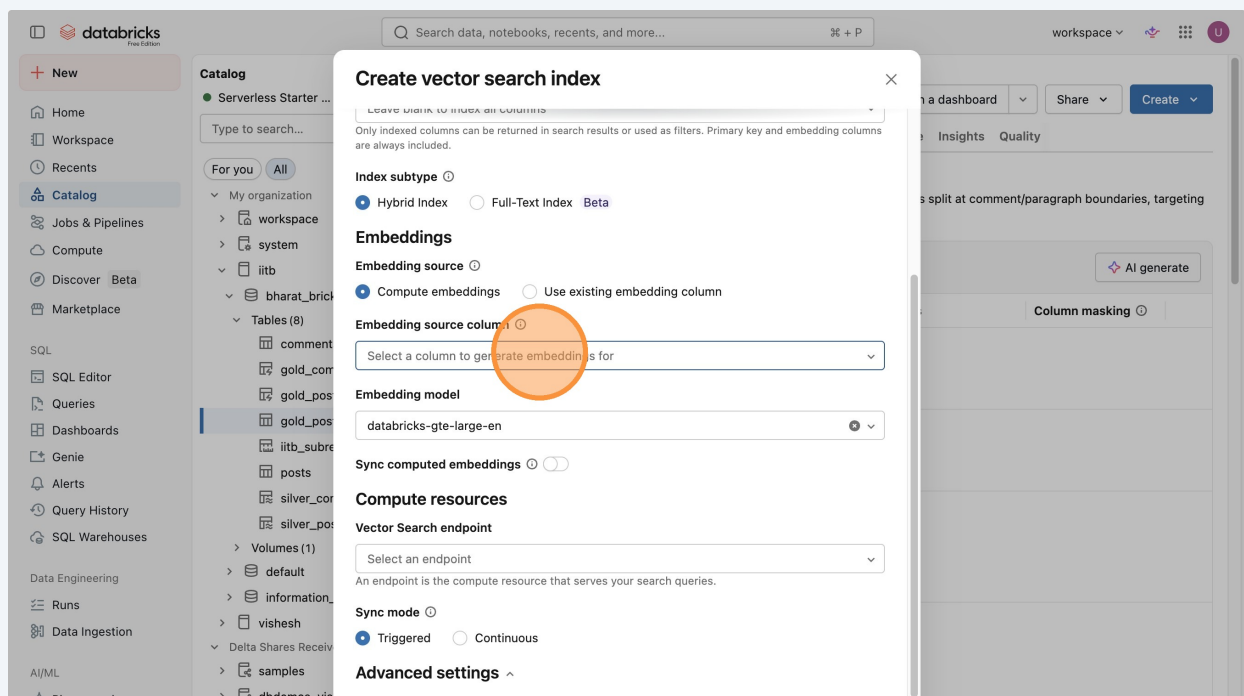
17 Click the "Select a primary key" field.



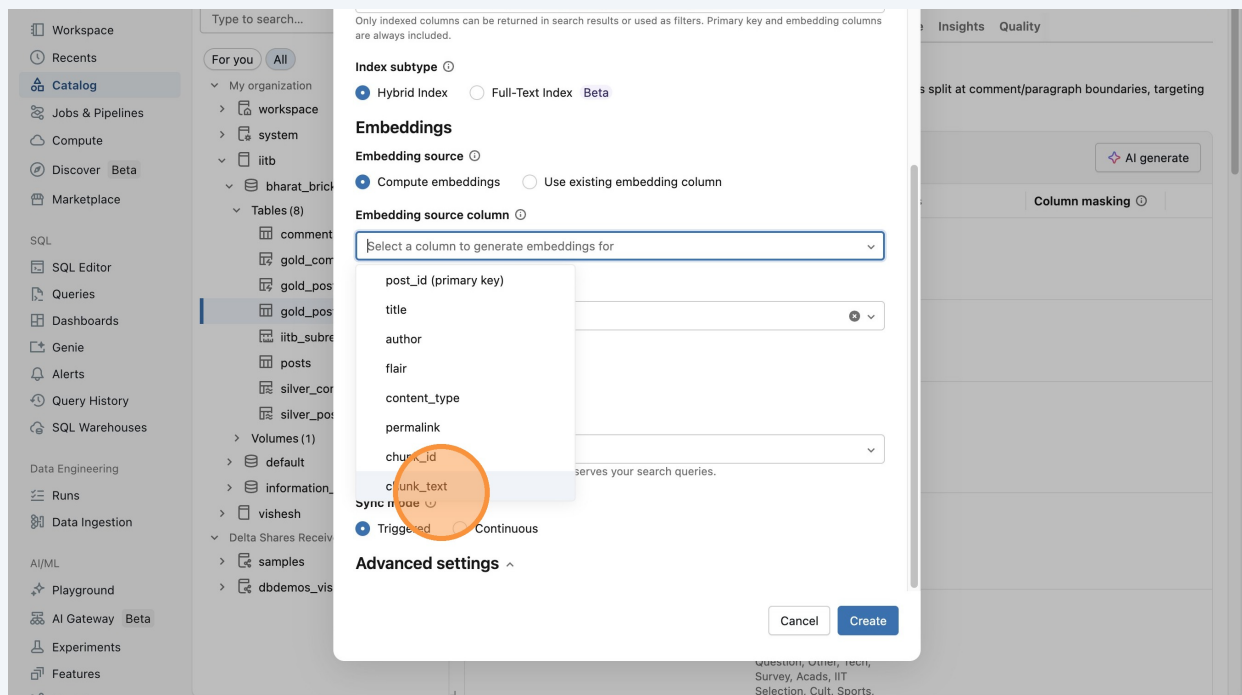
18 Click "post_id"



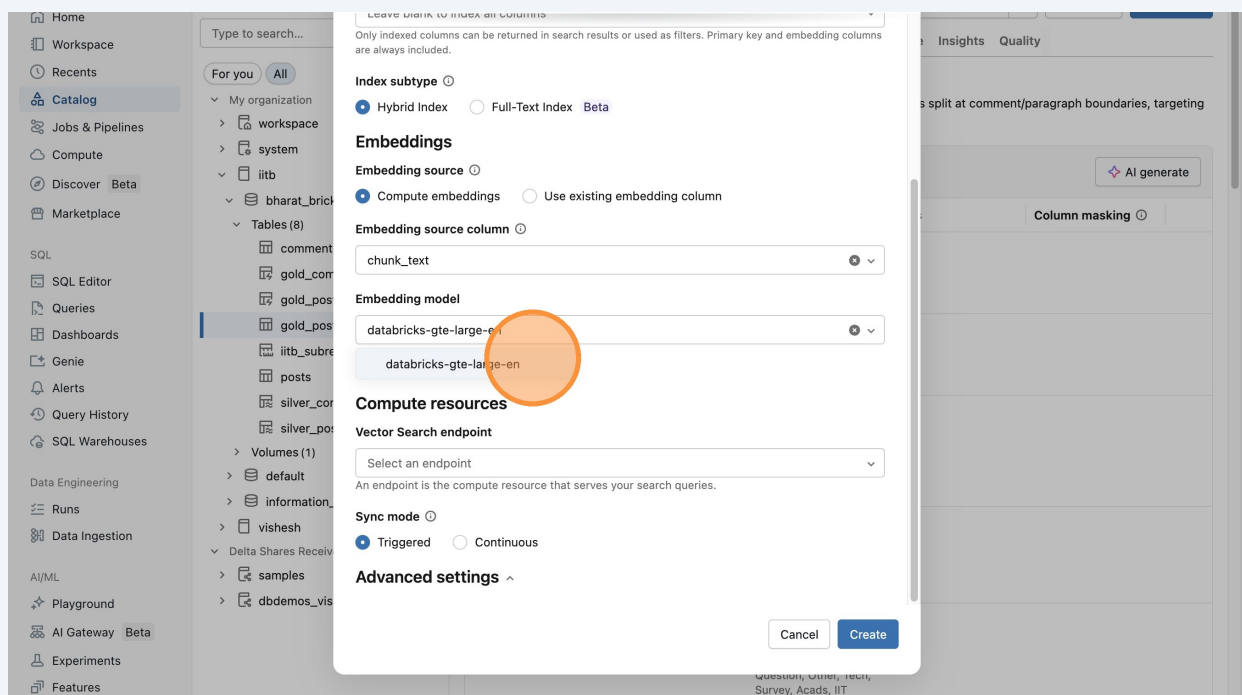
19 Click the "Select a column to generate embeddings for" field.



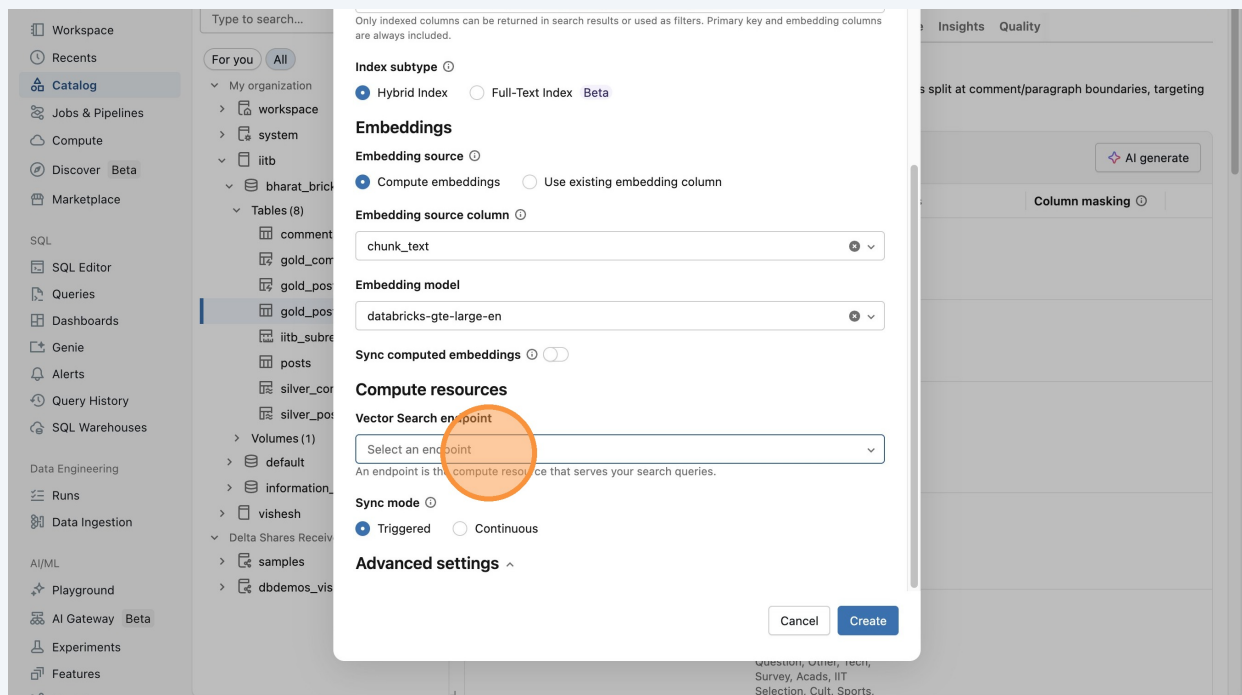
20 Click "chunk_text"



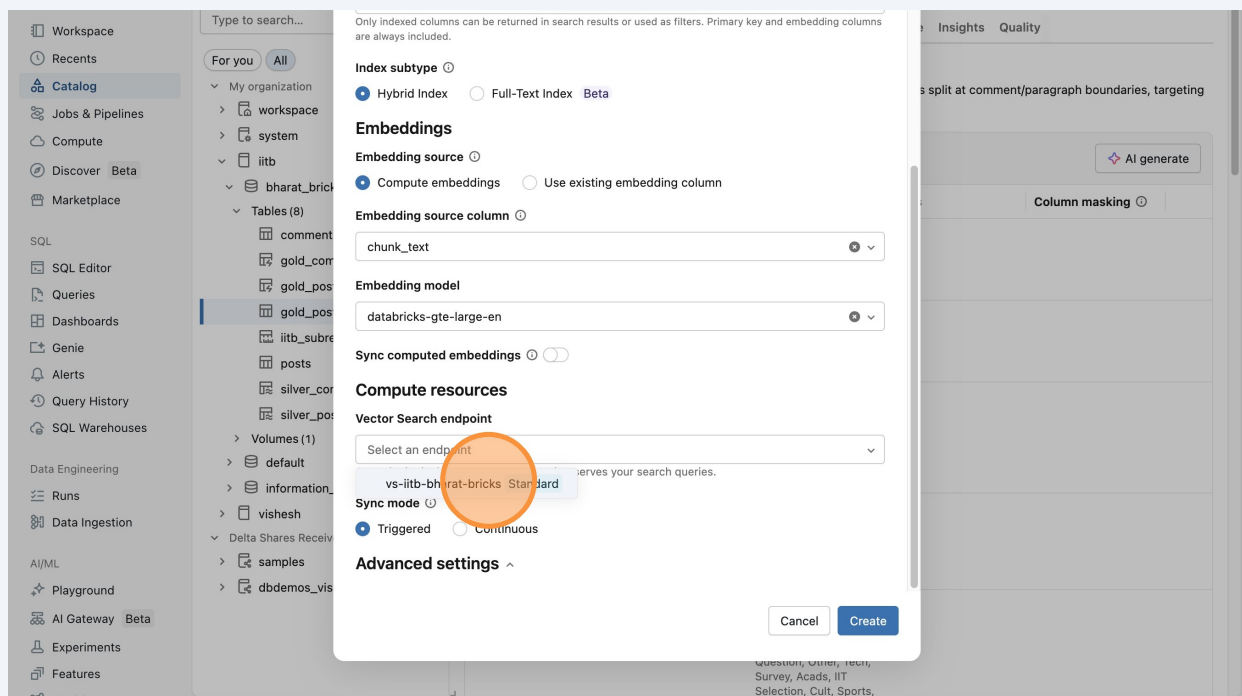
21 Click "databricks-gte-large-en"



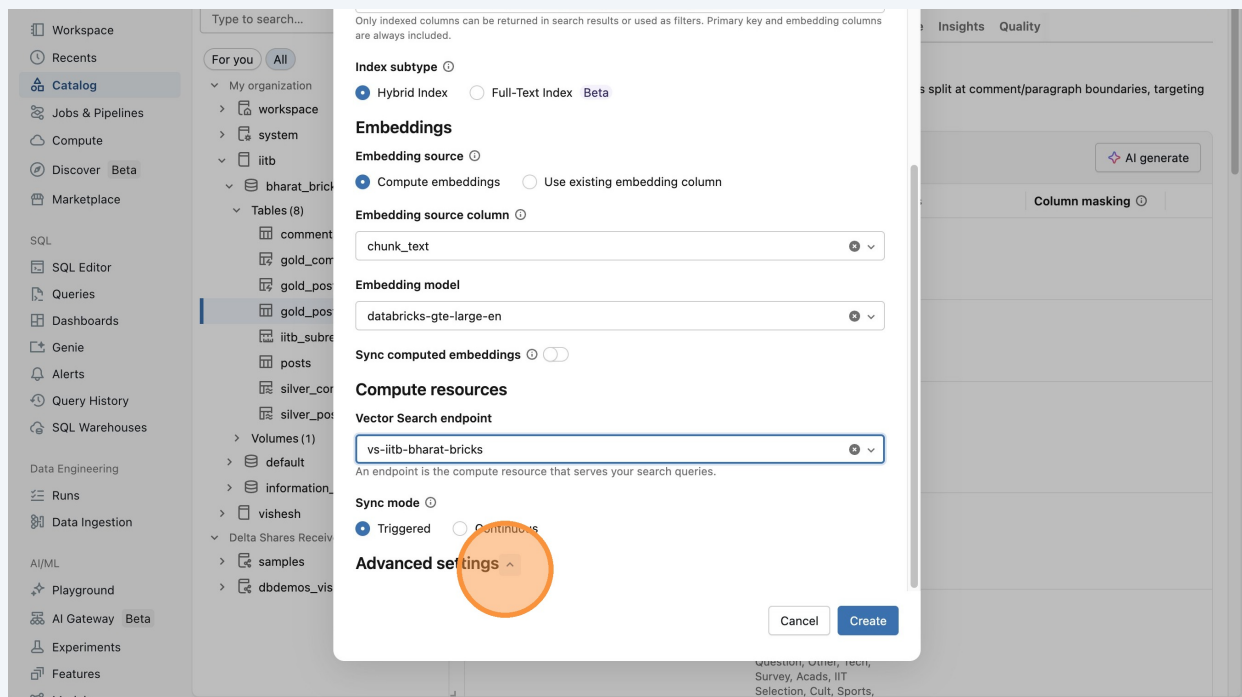
22 Click the "Select an endpoint" field.



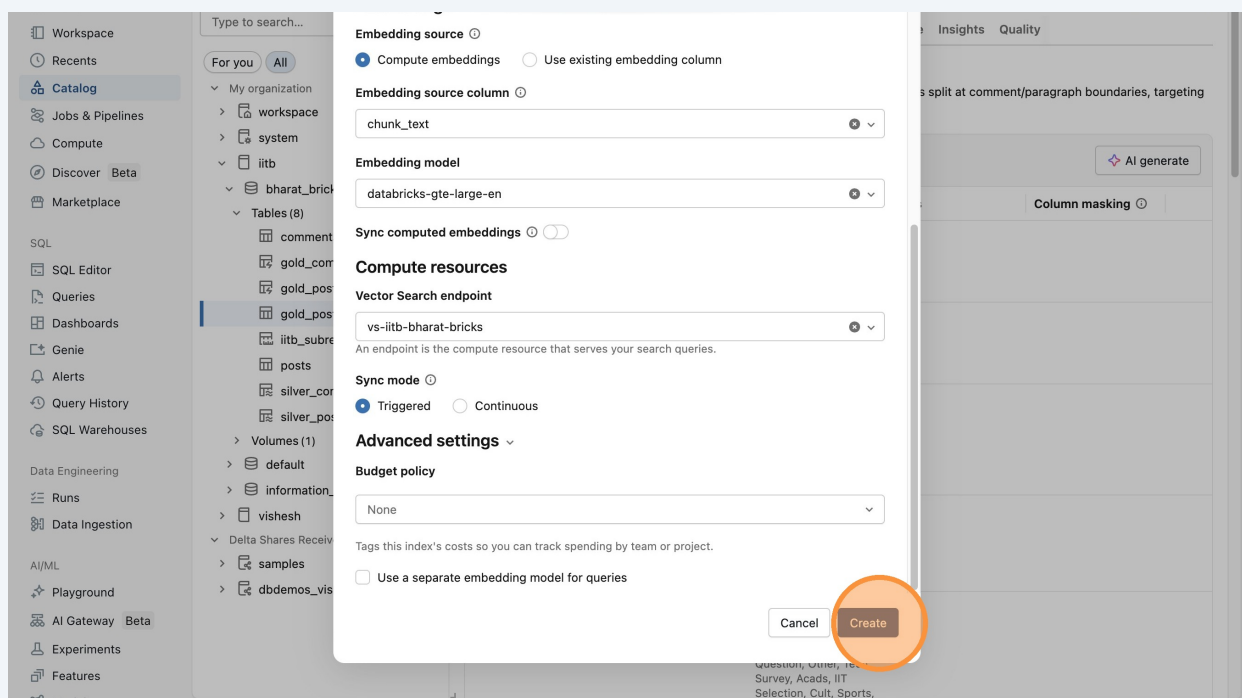
23 Click "vs-iitb-bharat-bricks Standard"



24 Click this icon.



25 Click "Create"



26 Click "vs_gold_posts_index"

The screenshot shows the Databricks workspace interface. On the left, the 'Catalog' sidebar is open, displaying a tree view of the workspace structure. The path is: My organization > workspace > system > iitb > bharat_bricks > Tables (9). The table 'vs_gold_posts_index' is highlighted with an orange circle. The main panel shows the 'Sample Data' tab for this table, displaying a list of rows with columns including '_db_chunk_text_vector'.

27 Click "_db_chunk_text_vector"

The screenshot shows the Databricks workspace interface. The 'Catalog' sidebar is open, and the path is: My organization > workspace > system > iitb > bharat_bricks > Tables (9). The table 'vs_gold_posts_index' is highlighted. The main panel shows the 'Sample Data' tab for this table, displaying a list of rows with columns including '_db_chunk_text_vector'. The column '_db_chunk_text_vector' is highlighted with an orange circle.